

Divyam Tyagi

Senior Backend Engineer – Scalable Systems

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SUMMARY

Backend engineer with **4+ years** of experience designing and scaling high-throughput backend systems using Python and FastAPI. Led a team of 7 engineers while building production-grade APIs, distributed systems, PostgreSQL-backed applications, Redis caching layers, and AI-powered RAG platforms. Strong expertise in asynchronous programming, system design, performance optimization, scalable microservices, and cloud-native backend development.

EXPERIENCE

Software Development Engineer II – Backend Lead

Irish Taylor

Sept 2023 – Present

Remote / India

- Led a team of **7 backend engineers** architecting a large-scale RAG platform, owning backend architecture, API design, code reviews, engineering standards, and delivery.
- Designed high-throughput asynchronous FastAPI microservices handling over **10,000 requests per day**, reducing **p95 API latency by 42%** through profiling, async optimizations, and database tuning.
- Built Redis multi-level caching, Celery background workers, Pub/Sub messaging, and intelligent rate limiting, enabling the platform to support over **4× traffic growth** without additional infrastructure.
- Optimized PostgreSQL performance using indexing strategies, query optimization, connection pooling, and execution-plan analysis, reducing slow-query execution time by **61%**.
- Collaborated with AI/ML teams to integrate production-ready LLM and Retrieval-Augmented Generation (RAG) pipelines using OpenAI, vector databases, and embedding services.

Independent Backend Consultant

Remote

May 2021 – Sept 2023

- Delivered scalable backend systems and REST APIs for over **6 clients** using Python and FastAPI, reducing manual operational effort by approximately **70%** for business workflow automation.
- Integrated payment gateways, CRM platforms, and third-party business APIs, deploying containerized applications using Docker.

PROJECTS

OmniData – High-Throughput Microservice

FastAPI, PostgreSQL, SQLAlchemy

[\[Repo\]](#)

- Built an asynchronous FastAPI microservice with dependency injection, optimized PostgreSQL indexing, and SQLAlchemy ORM architecture.
- Reduced average API response time by **48%** using Redis caching, efficient query planning, and asynchronous request processing.
- Implemented JWT authentication, structured logging, pagination, filtering, and production-ready API documentation.

Synapse – AI Document Intelligence

FastAPI, OpenAI, pgvector

[\[Repo\]](#)

- Built an AI-powered document intelligence platform capable of extracting, summarizing, classifying, and searching enterprise documents using embedding-based retrieval.
- Implemented RAG pipelines using vector embeddings, reranking, structured outputs, token usage monitoring, rate-limit handling, and asynchronous processing.

SKILLS

Languages	Python, JavaScript, SQL, Bash, C++
Backend	FastAPI, REST APIs, AsyncIO, Pydantic v2, SQLAlchemy 2.0, Dependency Injection, Web-Sockets, Server-Sent Events, Microservices
Databases	PostgreSQL, Redis, MongoDB, pgvector, Query Optimization, Indexing, Connection Pooling
Distributed Systems	Celery, Redis Pub/Sub, Background Workers, Rate Limiting, Caching, Event-Driven Architecture
Cloud & DevOps	Docker, Docker Compose, GitHub Actions, AWS EC2, AWS S3, ECS, Linux, Git
AI / LLM	OpenAI API, Gemini API, LangChain, LangGraph, RAG, Embeddings, Vector Search, Prompt Engineering, AI Agents, Agentic Workflows

EDUCATION

Bachelor of Technology (B.Tech.) – Computer Science & Engineering

Meerut Institute of Engineering and Technology (MIET), Meerut

2016 – 2020